

Deliverable 1

Structural Aliasing Analysis in Time Series Forecasting using Chronos

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This document must address the following key questions:

- What is the specific subject of study?
- Why is this subject important?
- Who is the intended audience?
- What impact should the project have on their understanding?
- How will the project achieve its communication objectives?
- Descrizione dettagliata dei diversi punti da affrontare
- Quali sono i risultati attesi?
- Quali metodologie utilizzare per costruire dati sintetici?
- ?Analisi rappresentatività? (perdita di informazione e problematiche embedding (Probing MDL????))
- Scelta di un dominio
- Analisi sui dati reali e risultati attesi (Bayesian statistics)
- Implicazioni in termini di sicurezza sul dominio

Possibile flusso logico:

- Comprendere le serie temporali è fondamentale in molteplici campi, tra questi ...
- L'analisi delle serie temporali è tradizionalmente affrontata mediante modelli autoregressivi, soluzioni basate su reti neurali rappresentano oggi una possibile alternativa.
- La possibilità di sfruttare modelli basati sull'architettura transformer, precedentemente trainati su grandi quantità di dati, rappresenta una prospettiva promettente. [1]
- Descrizione di Chronos e delle sue varianti [2, 3, 4, 4]
- ??altri modelli alternativi?? [5]
- Questo studio ha l'obiettivo di analizzare il fenomeno dello structural aliasing di serie temporali che si può verificare utilizzando Chronos-Bolt (Chronos2). [6]
- Tuttavia, è necessario comprendere la capacità rappresentativa di questi modelli. [7]

References

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During the preparation of this work, the authors used AI-based tools, including GPT-5.5, GPT-5.4, Gemini 3.1 Pro, and Sonnet 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

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- [7] Elena Voita and Ivan Titov. “Information-Theoretic Probing with Minimum Description Length”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. EMNLP 2020. Ed. by Bonnie Webber et al. Online: Association for Computational Linguistics, Nov. 2020, pp. 183–196. DOI: 10.18653/v1/2020.emnlp-main.14. URL: <https://aclanthology.org/2020.emnlp-main.14/> (visited on 05/13/2026).